

Grain Boundaries Revisited: A Anisotropic Perspective

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Abstract

Work in materials research has long relied on assumptions we test directly here. Calibration drift over the trial period remained below the threshold set out in the protocol. The reduced model retains most of the explanatory power at a fraction of the computational cost. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

1. Introduction

We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. Residual error grows approximately linearly with the number of coupled parameters. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Longer observation windows narrow the confidence interval but do not shift the central estimate. The reduced model retains most of the explanatory power at a fraction of the computational cost. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. Neither of the alternative formulations improved on the simple additive model in our setting. Discrepancies at the boundary are consistent with the simplifying assumptions noted above.

The measurements reported here were collected over twelve independent trials under controlled conditions. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Neither of the alternative formulations improved on the simple additive model in our setting. The reduced model retains most of the explanatory power at a fraction of the computational cost.

2. Related Work

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threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification.

3. Materials and Methods

Our baseline reproduces the behaviour described in earlier work to within the stated tolerance. The reduced model retains most of the explanatory power at a fraction of the computational cost. Residual error grows approximately linearly with the number of coupled parameters. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Sensitivity to the smoothing window proved modest across the range we examined. Neither of the alternative formulations improved on the simple additive model in our setting.

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4. Experimental Setup

Neither of the alternative formulations improved on the simple additive model in our setting. Longer observation windows narrow the confidence interval but do not shift the central estimate. Residual error grows approximately linearly with the number of coupled parameters. The measurements reported here were collected over twelve independent trials under controlled conditions. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

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5. Results

Neither of the alternative formulations improved on the simple additive model in our setting. Discrepancies at the boundary are consistent with the simplifying assumptions noted above. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. The measurements reported here were collected over twelve independent trials under controlled conditions. Taken together, these findings argue for treating the two regimes separately in future analyses. Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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6. Discussion

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7. Limitations

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